

## POWER MARKETS

# Volume II Q&A Overview: Commercial Analysis, Contracts, And Asset Decisions

The second volume follows the path from physical delivery to project economics: cash-flow models, PPA structures, storage choices, dispatch, valuation, portfolio risk, and partner governance.

# How does the system become a priced decision?

|           |                                                                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | What converts a physical project into a commercial recommendation?                                                                                      |
| ANSWER    | A project becomes investable when resource, grid, market, contract, finance, and operating-control assumptions can be connected in one cash-flow logic. |
| DEEPEN IN | Vol II Ch 1, Ch 2, Ch 3, Ch 13, Ch 16.                                                                                                                  |

1

## Physical profile

- Weather, availability, degradation, curtailment, and technology behavior define hourly production.

2

## Market conversion

- Capture price, congestion, balancing exposure, tariffs, losses, and liquidity convert output into market value.

3

## Contract structure

- Price formula, quantity promise, settlement basis, certificates, credit, tenor, and fallback rules allocate risk.

4

## Financing view

- Debt sizing, downside cases, contracted revenue quality, merchant tail, and liquidity decide bankability.

5

## Operating control

- Forecasts, dispatch rights, BRP/BSP execution, data rights, and risk limits decide whether the promise can be governed.

# What belongs in the project cash-flow path?

|           |                                                                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | Which assumptions connect engineering, market, contract, and finance views?                                                            |
| ANSWER    | The model has to preserve the path from physical performance to realized cash, otherwise it hides where value or risk actually enters. |
| RELATION  | Project cash flow starts from delivered MWh × realized price, then subtracts grid, operating, financing, and risk costs.               |
| DEEPEN IN | Vol II Ch 1, Ch 3, Ch 13.                                                                                                              |

1

## Physical input

- Hourly generation, degradation, outages, curtailment, connection capacity, and operating limits.

2

## Price realization

- Capture, basis, congestion, tariffs, losses, imbalance, and product eligibility.

3

## Contract conversion

- Fixed prices, indexation, floors, collars, certificate treatment, settlement basis, tenor, and residual exposure.

4

## Financing conversion

- Debt case, merchant tail, downside sensitivities, counterparty risk, collateral, and covenant headroom.

# Why can grid access outrank resource quality?

**QUESTION** How do connection and tariff terms enter project economics?

**ANSWER** Grid access is a priced operating right. It defines what the asset can inject, withdraw, flex, or reserve at the location.

**RELATION** Capacity charges scale with MW rights; usage charges and losses scale with MWh flows.

**DEEPEN IN** Vol I Ch 4; Vol II Ch 2; Case C03.

1

## Connection cost and timing

- Upfront cost, reinforcement delay, and queue risk change sequencing and hurdle rates.
- A high-resource site can lose if the grid right arrives late or remains uncertain.

2

## Capacity charges

- MW-based charges can penalize rarely used import or export capacity.
- Co-located storage needs a connection design aligned with actual operation.

3

## Variable network cost

- MWh charges, losses, and curtailment compensation affect dispatch and revenue.
- The relevant cost can depend on import, export, time, voltage level, or network boundary.

4

## Conditional access

- A smaller or interruptible right can be attractive when the operating model can manage the constraint.

# How does renewable shape become price?

|           |                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | Why can a project with low cost and high annual generation have weak realized revenue?                                   |
| ANSWER    | Renewable value follows production-hour value. Correlation, curtailment, and negative prices enter through hourly shape. |
| RELATION  | Capture price = $\text{sum}(\text{price}_t \times \text{generation}_t) / \text{sum}(\text{generation}_t)$ .              |
| DEEPEN IN | Vol I Ch 12; Vol II Ch 3, Ch 6.                                                                                          |

## Capture price

- Generation-weighted price measures the price earned by the asset's production shape.
- Solar capture can fall when solar output is synchronized across the zone.

## Curtailment and negative prices

- Curtailment removes MWh from the revenue stream.
- Negative prices can make production rights, subsidies, contracts, and curtailment logic commercially important.

## Portfolio effect

- A portfolio model needs correlation across assets, zones, contracts, and customer obligations.
- Diversification is valuable when it changes hard hours and portfolio exposure.

## Modeling implication

- Hourly simulation is the normal working level for capture, shaping, and imbalance.
- Annual summaries are outputs; hourly profiles are the working inputs.

# What is a PPA allocating?

QUESTION

ANSWER

DEEPEN IN

Why is the label "PPA" less important than the risk allocation?

A power purchase agreement assigns mismatch: quantity, timing, settlement reference, certificates, control, credit, fallback, and termination.

Vol II Ch 4, Ch 5; Vol III Ch 5; Case C01.

## Risk-allocation map

| Dimension    | Commercial question                      | Seller exposure                 | Buyer value               |
|--------------|------------------------------------------|---------------------------------|---------------------------|
| Quantity     | As-produced, fixed, shaped, as-consumed? | Shortfall, surplus, balancing   | Usable energy shape       |
| Settlement   | Physical or financial, which reference?  | Basis and liquidity mismatch    | Procurement fit           |
| Price        | Fixed, indexed, floor, collar?           | Merchant upside/downside        | Budget certainty          |
| Certificates | Bundled, unbundled, granular?            | Evidence burden                 | Claim quality             |
| Control      | Who curtails, dispatches, nominates?     | Operational constraint          | Reliability of promise    |
| Credit       | Collateral and termination rights?       | Liquidity and counterparty risk | Deliverability confidence |

# What turns a PPA idea into a quote?

|           |                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How does a seller move from product language to a defensible price?                                                              |
| ANSWER    | A quote needs to show every transformation from raw production value to customer promise, residual exposure, credit, and margin. |
| RELATION  | Indicative quote = raw energy + shaping/firming + imbalance/grid/operations + credit/optionality + margin.                       |
| DEEPEN IN | Vol II Ch 6, Ch 7; Case C01.                                                                                                     |

1

## Raw energy

- Expected production value before shaping, firming, or certificate treatment.

2

## Shape and firming

- Cost of converting production into the customer's profile through portfolio netting, storage, market purchase, or fallback.

3

## Imbalance, grid, and operations

- Forecast error, settlement exposure, tariffs, losses, curtailment, data handling, and partner execution.

4

## Credit and optionality

- Collateral, tenor, termination rights, merchant tail, retained flexibility, and residual uncertainty.

5

## Margin and refusal boundary

- The offer boundary is reached when residual risk or control-right gaps exceed what the price can carry.

# When does a sales promise become investable?

**QUESTION** What has to be true before a customer product supports capital allocation?

**ANSWER** The product has to be backed by assets, market access, contracts, data, operating rights, and downside cases that make the promise governable after signing.

**DEEPEN IN** Vol II Ch 7, Ch 13, Ch 16; Cases C01, C07.

1

## Promise

- Product shape, settlement basis, certificates, fallback rules, reporting, and tolerance bands.

2

## Support

- Existing portfolio, new build, BESS, market procurement, partner supply, or demand flexibility.

3

## Model

- Hourly delivery, residual exposure, downside cases, liquidity need, and sensitivity to key assumptions.

4

## Execution

- Dispatch rights, BRP/BSP responsibilities, data rights, risk limits, and customer reporting obligations.

5

## Approval

- Investment committee needs the binding constraint, evidence gap, owner, and assumption that would flip the answer.



# What does a BESS revenue stack spend?

|           |                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | Why can apparently additive battery revenues conflict?                                                                                       |
| ANSWER    | Each revenue line claims the same scarce asset capability: MW, MWh, state of charge, cycles, grid rights, qualification, and control rights. |
| ANCHOR    | A 50 MW battery cannot simultaneously sell the same 50 MW into every attractive product; reservation and opportunity cost come first.        |
| DEEPEN IN | Vol I Ch 13; Vol II Ch 8, Ch 10, Ch 12; Case C04.                                                                                            |

1

## Energy shifting

- Gross spread has to survive efficiency loss, fees, degradation, and grid costs.

2

## Reserve capacity

- MW held available may require SoC headroom, sustain capability, telemetry, and activation readiness.

3

## Co-location

- Storage can reduce curtailment, share grid rights, shape a PPA, or change export/import constraints.

4

## Customer product

- A shaped or firmed promise can reserve capacity that would otherwise chase market revenue.

5

## Revenue quality

- Contracted floors, regulated services, merchant arbitrage, optimizer upside, and strategic optionality carry different financing weights.

# When is extra duration worth buying?

|           |                                                                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How are 1h, 2h, and 4h storage choices compared?                                                                                                               |
| ANSWER    | Extra MWh is a marginal investment. It is attractive only when the incremental scarcity hours, product depth, and operating limits clear the incremental cost. |
| RELATION  | Extra-duration hurdle = annualized cost of added MWh + opportunity cost, compared with incremental net revenue in the marginal hours.                          |
| DEEPEN IN | Vol II Ch 9, Ch 12; Vol III Ch 6; Case C02.                                                                                                                    |

1

## Identify the marginal hours

- Which hours require the additional MWh after the first hour or two are exhausted or reserved?

2

## Price the incremental asset

- Add capex, augmentation, degradation, insurance, warranty, tariff impact, and opportunity cost.

3

## Test product depth

- A real revenue product can still be too shallow for the amount of capacity entering the market.

4

## Check substitutes

- Portfolio diversity, demand flexibility, contract design, grid rights, or waiting can beat extra duration.

# What exactly did the optimizer solve?

|           |                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How should a dispatch result be interpreted when the model has limited information and rights?                                     |
| ANSWER    | The dispatch schedule is the visible output of a specified objective, constraint set, information set, and control-right boundary. |
| RELATION  | Maximize value subject to MW, MWh, state-of-charge, grid, contract, degradation, and risk constraints.                             |
| DEEPEN IN | Vol II Ch 10, Ch 12, Ch 15; Cases C04, C05.                                                                                        |

## Model specification

| Layer      | Example variable      | Binding constraint               | Output to inspect  |
|------------|-----------------------|----------------------------------|--------------------|
| Energy     | Charge and discharge  | MW, MWh, efficiency, SoC         | Energy schedule    |
| Reserve    | Committed MW          | Sustain, headroom, qualification | Capacity offer     |
| Contract   | PPA delivery          | Shape, penalties, control rights | Customer exposure  |
| Asset life | Throughput and cycles | Degradation, warranty, insurance | Net margin         |
| Grid       | Import and export     | Connection and tariff limits     | Feasible operation |
| Risk       | Position limits       | Liquidity, credit, governance    | Allowed dispatch   |

# When does a forecast have value?

|           |                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------|
| QUESTION  | Why is forecast accuracy insufficient as a commercial metric?                                           |
| ANSWER    | A forecast earns money only when it changes a feasible action before the relevant market window closes. |
| RELATION  | Forecast value depends on error reduction × exposed volume × actionability before gate closure.         |
| DEEPEN IN | Vol I Ch 8; Vol II Ch 11; Vol III Ch 2, Ch 9.                                                           |

1

## Error distribution

- Production uncertainty affects schedules, imbalance exposure, dispatch, and PPA delivery risk.

2

## Decision window

- The value of the forecast depends on gate closure, liquidity, market access, and control rights.

3

## Action

- The response can be a trade, nomination change, battery dispatch, reserve offer, hedge, or customer-risk adjustment.

4

## Attribution

- Forecast quality needs benchmark comparison, realized PnL attribution, and records of when the model was allowed to act.

# How do warranty and insurance enter dispatch?

|           |                                                                                                                                                      |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | Why can a profitable-looking battery action be infeasible?                                                                                           |
| ANSWER    | Asset-life constraints are commercial constraints: each dispatch action can consume part of a warranty, augmentation budget, or insurance allowance. |
| ANCHOR    | Cycles, throughput, state-of-charge windows, and insurance limits can bind before the apparent market spread is fully captured.                      |
| DEEPEN IN | Vol II Ch 12; Vol II Ch 8, Ch 10; Cases C02, C04.                                                                                                    |

1

## Throughput and cycles

- Revenue consumes cycle budget, equivalent full cycles, or throughput allowances.

2

## Operating envelope

- C-rate, SoC window, temperature, safety policy, and revenue mode can constrain dispatch.

3

## Contract interaction

- A revenue line that looks attractive in gross terms can violate cycle caps, warranty assumptions, or insurance policy limits.

4

## Governance implication

- The dispatch model needs an explicit cost or constraint for asset-life consumption.

# Why can the same expected value finance differently?

|           |                                                                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | Why does expected value fail as a complete investment metric?                                                                                      |
| ANSWER    | Contracted revenue, merchant upside, liquidity risk, downside exposure, and counterparty quality have different uses in debt and equity decisions. |
| DEEPEN IN | Vol II Ch 13, Ch 17; Vol II Ch 6.                                                                                                                  |

## Bankable quality

- Fixed or floor-like revenues can support debt when obligations, counterparty risk, and delivery exposure are controlled.
- Merchant tail has value but usually less debt capacity.

## Downside sensitivity

- Capture erosion, curtailment, tariff changes, reserve saturation, imbalance exposure, and collateral calls move the debt and equity case differently.
- The model identifies which lever changes value: resource, grid, contract, storage, partner, financing, or portfolio context.

# What does portfolio control measure?

|           |                                                                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How does the risk view change once projects, PPAs, storage, and partners interact?                                                                             |
| ANSWER    | A portfolio is a correlation system with liquidity and governance constraints. The risk pack has to show shape, basis, volume, credit, and execution together. |
| DEEPEN IN | Vol II Ch 14, Ch 17; Vol III Ch 12; Case C07.                                                                                                                  |

## Portfolio risk map

| Risk      | Measurement               | Control question               | Failure mode                 |
|-----------|---------------------------|--------------------------------|------------------------------|
| Shape     | Capture and residual load | Which hours drive value?       | Comfort from annual averages |
| Basis     | Zone/reference spread     | Where does settlement diverge? | Unpriced location exposure   |
| Volume    | Weather and curtailment   | Who owns shortfall?            | Contract overpromise         |
| Credit    | Counterparty exposure     | What collateral is required?   | Liquidity squeeze            |
| Execution | Benchmark leakage         | What was feasible?             | Outsourced opacity           |

# How is outsourced execution governed?

|           |                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | What does an asset owner need from a BRP/BSP or optimizer partner?                                                               |
| ANSWER    | Outsourced execution still needs asset-owner observability: data rights, feasible benchmarks, attribution, and escalation rules. |
| ANCHOR    | A partner scorecard is credible only when it uses the information, rights, liquidity, and constraints available at the time.     |
| DEEPEN IN | Vol II Ch 15, Ch 16, Ch 17; Vol III Ch 10; Case C05.                                                                             |

1

## Data rights

- Forecasts, bids, schedules, activations, metering, settlement, SoC, outages, and constraints need useful resolution.

2

## Feasible benchmark

- The counterfactual has to respect information timing, control rights, liquidity, and operational constraints.

3

## Attribution

- Underperformance can come from market conditions, forecast error, poor bidding, conservative limits, data problems, or incentives.

4

## Escalation

- Scorecards work when materiality, evidence, contract rights, and escalation paths are connected.